Bayesian Networks and Markov Models in Action

Project Instructions

Autumn 2026

1 Goal

With your partner, you will:

1. **Choose a topic**: a question that can be answered with data, real or simulated, and whose answer is uncertain;
2. **Model it**: write a Bayesian model for the data, with its unknowns, its likelihood (or simulator) and its priors;
3. **Implement** a simulation-based method of the course to approximate the posterior;
4. **Analyse the results**: check that the method works, interpret the posterior, and answer the question with its uncertainty.

The project assesses the implementation, the simulation, and the analysis. A simple model that is carefully implemented, checked and interpreted is worth more than an ambitious model whose output cannot be trusted.

2 Requirements

Your project must meet the following requirements.

1. A Bayesian model, fully specified.
State the question, the data, the unknown parameters, and latent variables, if any, the likelihood or the simulator, and the priors with a justification for each. A diagram of the model as a Bayesian network is encouraged. Define every symbol you use.

2. A simulation method that you implement yourselves.
Use at least one method of the course: accept–reject, importance sampling, a Monte Carlo optimization method (simulated annealing, EM, stochastic approximation), Metropolis–Hastings, Gibbs sampling, or approximate Bayesian computation. Write the sampler or the algorithm yourselves, in Python. Libraries are welcome for random number generation, linear algebra, data handling and plots (NumPy, SciPy, pandas, Matplotlib). Probabilistic programming libraries (PyMC, Stan, NumPyro, ...) may be used *only* to cross-check your own results, never to replace them.

3. Validation on simulated data.
Before using real or realistic data, simulate a data set from your model with parameter values that you choose, run your method on it, and show that it recovers these values: the true values should fall within the posterior intervals. This is the only way to separate a bug from a surprising result.

4. Convergence and accuracy diagnostics.
Show that the numbers can be trusted, with the tools that fit your method: trace plots, several chains from dispersed starting points and $\hat{R}$, autocorrelations and effective sample sizes, acceptance rates, Monte Carlo standard errors, effective sample size of importance weights, or, for ABC, the effect of the tolerance and the coverage of the intervals.

Beyond these requirements, a good analysis reports posterior summaries with their uncertainty (e.g., means, intervals, probabilities of events of interest), checks the fit of the model with posterior predictive simulations, studies the sensitivity of the conclusions to the priors, and discusses the limitations of the model.

3 The pitch: Friday 16 October 2026

Each group presents the general idea of its project in **5 minutes**. The pitch is not graded and its purpose is to validate your topic. Prepare at most three slides that answer:

1. What is the question, and why does it matter?
2. What data will you use (e.g., real, public or simulated), and what do you observe?
3. What is the model: unknowns, likelihood or simulator, priors?
4. Which method do you plan to implement, and what do you expect to be difficult?

4 The report

- Use the LaTeXtemplate disponible in EduNao.
- At most **8 single-column pages**, excluding the references and the appendices. The appendices may contain derivations, additional figures and diagnostics, but the main text must be understandable without them.
- A suggested structure includes:
    1. *Introduction*: the question, its context, and what you found.
    2. *Data*: source, size, preprocessing, and a descriptive figure.
    3. *Model*: the Bayesian model, with every symbol defined and every prior justified.
    4. *Method*: the algorithm, its tuning choices (proposals, step sizes, number of iterations, tolerances) and their justification.
    5. *Validation*: recovery of known parameters on simulated data.
    6. *Results*: posterior summaries, diagnostics, posterior predictive checks, sensitivity analysis.
    7. *Discussion*: the answer to the question, the limitations, and what you would do next.
- Every number in the report must be produced by the submitted code. Figures must have labeled axes and a caption that explains what they show.
- Cite all sources: data, articles, books, and any code that you reuse or adapt.

5 The code

- Python, as Jupyter notebooks or scripts. The course environment (`environment.yml`) is recommended; list any additional package in a `requirements.txt` file.
- A `README.md` file that explains how to run the code and which file produces which figure or table of the report.
- Fixed random seeds, so that running the code reproduces the numbers of the report.
- The data, or a script that downloads it if it is large or cannot be redistributed.
- The code must run from start to end without manual intervention. Indicate its running time if it exceeds a few minutes.

6 Submission

One member of the group uploads a single zip archive on EduNao by **Monday 16 November 2026, 23:59** (Paris time). Name the archive `project-<surname1>-<surname2>.zip`, with the surnames of both members. It contains:

- `report.pdf`, the report;
- `report/`, the LaTeX sources of the report;
- `code/`, the notebooks or scripts, the data (or a download script), the `README.md` and, if needed, the `requirements.txt` file.

7 Grading

The report and the code are graded together, with the following rubric.

Criterion	Weight	What we look for
Modeling	25%	a clear question; a model that fits it; every term defined; priors justified; assumptions stated
Implementation	25%	a correct method, implemented by the group; sensible tuning; readable and reproducible code
Results and diagnostics	30%	validation on simulated data; convergence and accuracy diagnostics; posterior summaries with uncertainty; predictive checks; sensitivity; an answer to the question
Report	20%	structure and clarity; quality of figures and tables; honest discussion of the limitations; respect of the page limit

Both members of a group receive the same grade, and both must be able to explain any part of the project.

8 Example of topics

The topics below are suggestions; you may propose your own. Several are extensions of examples, labs and exercises of the course. When you use public data, check its license and cite it.

T1. A/B tests across many campaigns. A company runs the same test in twenty countries with very different traffic. Which variant is best, in which country? A hierarchical Beta–Binomial or logistic model that shares information between countries. *Methods*: Gibbs or Metropolis-within-Gibbs; the probability that each variant is best and the expected loss of each decision by Monte Carlo.

T2. Customer lifetime value. How many purchases will each customer make next year, and which customers are already lost? A buy-till-you-die model (Pareto/NBD or BG/NBD), where each customer has a latent purchase rate and a latent lifetime. *Methods*: Gibbs sampling with data augmentation on the latent lifetimes. *Data*: transaction logs, such as the CDNOW data set used in the customer-base literature.

T3. Volatility of financial returns. How does the volatility of a stock index evolve, and what is the probability of a daily loss beyond 3%? A stochastic volatility model, in which the log-volatility is a latent autoregressive Markov chain. *Methods*: Metropolis-within-Gibbs on the latent volatilities, or particle methods. *Data*: public daily prices of an index.

T4. Rare losses and risk measures. Estimate the value-at-risk and the expected shortfall of a portfolio with heavy-tailed returns at the 99.9% level. *Methods*: plain Monte Carlo against importance sampling for the rare event, with a careful study of the variance of each estimator; posterior uncertainty on the tail parameters by Metropolis–Hastings.

T5. Credit scoring. What is the probability that a loan applicant defaults, and how uncertain is it? A Bayesian logistic regression, possibly hierarchical across loan types. *Methods*: Metropolis–Hastings with a tuned proposal covariance, against importance sampling with a Laplace proposal. *Data*: a public credit data set.

T6. Ranking hospitals. Which hospitals have an unusually high mortality after a given surgery, once the small number of operations in some of them is taken into account? A hierarchical binomial model with shrinkage, and the posterior distribution of each hospital's rank. *Methods*: Gibbs or Metropolis-within-Gibbs. *Data*: published surgical outcome data, or data simulated to resemble them.

T7. An epidemic from daily case counts. What are the transmission and recovery rates of an outbreak, and when will it peak? A stochastic SIR model, whose likelihood is intractable when only the daily new cases are observed. *Methods*: ABC rejection and ABC–SMC with well-chosen summary statistics, validated by coverage. *Data*: simulated, or public surveillance counts.

T8. An emergency department queue. How long do patients wait, and how many more doctors would halve the waiting time? A queueing simulator with arrival and treatment rates, observed only through arrival and departure times (an extension of Lab 7). *Methods*: ABC, then posterior predictive simulation of the waiting times under alternative staffing.

T9. A change in the rate of events. Did the rate of customer complaints, website failures or accidents change after an intervention, and when? A Poisson model with an unknown change point. *Methods*: Gibbs sampling, with the change point as a discrete parameter; comparison with the model without a change.

T10. Diagnosis with a Bayesian network. Given some symptoms and test results, what is the probability of each disease, when some variables are missing? A small Bayesian network whose conditional probability tables are learned from incomplete data. *Methods*: Gibbs sampling over the missing values, or EM; comparison with exact inference on the small network.

T11. Word-of-mouth diffusion on a network. How strong is word of mouth for a product launch, and how fast does interest fade? An agent-based diffusion simulator on a social network (an extension of the referral campaign of Chapter 7). *Methods*: ABC, with a comparison of summary statistics and of the rejection and sequential versions.